

Population and Economic Growth

When Malthus Meets Romer

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One of the Oldest Debate in Economics

- Does a larger population **raise** or **lower** living standards?
 - **Malthus (1798)**: population strains resources $\Rightarrow$ wages to subsistence
 $\Rightarrow$ David Ricardo, Paul Ehrlich, William Catton, ...
 - **Romer (1990)**: more people $\Rightarrow$ more ideas $\Rightarrow$ innovation dominates
 $\Rightarrow$ Richard Jones, Julian Simon, Chad Jones, Elon Musk, ...
- Never more consequential: 2bn $\rightarrow$ 8bn+ in a century, now fertility below replacement across the developed world
- This paper: study the implications of the **disagreement itself**
- Channel: the co-evolution of belief distribution and population dynamics

This Paper

1. **Model**: OLG + Barro–Becker altruism + semi-endogenous growth + heterogeneous beliefs about scale effects (Romerians vs. Malthusians)
2. **Theory**: fertility ranking between types is **state-dependent**; wrong beliefs generate distinct pathologies:
 - Malthusian world + Romerians $\Rightarrow$ **boom-bust cycles**
 - Romerian world + Malthusians $\Rightarrow$ **dormant stagnation trap**
3. **Evidence**: Malthusian beliefs correlated with 0.3 fewer children (GSS) + Malthus at Haileybury (Robertson 2026)
4. **History**: the role of beliefs in demographic transitions
5. **Policy**: announcements as **unconventional family policy**

Related Literature & This Paper's Contribution

- **Fertility & unified growth:** Barro–Becker (1989), Galor (2005, 2011), Hansen–Prescott (2002); captures transition from stagnation to growth
- **Scale effects:** Romer (1990), Kremer (1993), Jones (1995, 2022); the sign of θ is central to long-run projections
- **Heterogeneous beliefs:** Angeletos–La’O (2013), Coibion–Gorodnichenko (2015); climate disagreement (Bakkensen–Barrage 2022)
- **Cultural transmission:** Bisin–Verdier (2001, 2011), Doepke–Zilibotti (2017); how worldviews persist across generations

This paper: disagreement about the population scale effect in a unified growth model where the error itself can hide its own evidence

Roadmap

- Model
- Empirical evidence: beliefs and fertility
- Long-run dynamics
- Demographic history interpretation
- Policy: the value of announcements

Model: Environment

- Discrete-time OLG; continuum of atomistic households
- Time endowment 1; fertility n costs time $\psi(n)$, labor supply $1 - \psi(n)$
- Linear production: $Y_t = A_t L_t$, wage = A_t , so

$$c_{j,t} = A_t(1 - \psi(n_{j,t}))$$

- A_t = “effective” TFP (incorporates congestion/dilution)
- Population: $N_{t+1} = \bar{n}_t N_t$, $\bar{n}_t = \phi_t n_{R,t} + (1 - \phi_t) n_{M,t}$

- Central state variable: **population gap** x_t = population relative to knowledge-consistent reference

$$\ln A_{t+1} = \ln A_t + g(x_t; \theta^*) + \eta_{t+1}, \quad x_{t+1} = (1 - \mu)x_t + \ln \bar{n}_t$$

- Leading case: $g(x; \theta) = \bar{g} + \theta x$, $\theta_R > 0 > \theta_M$
- μ = **catch-up rate** of the knowledge stock (semi-endogenous, Jones 1995):
 - one-time population increase $\Rightarrow$ **temporary** growth effect
 - sustained growth $\bar{n} > 1 \Rightarrow$ permanent: $g_\infty = g(\ln \bar{n} / \mu; \theta^*)$
- Disagreement is about the **sign** of $\theta^* \in \{\theta_R, \theta_M\}$

Key Structural Property: The Uninformative Reference

- Rival models **cross** at $x = 0$: $g(0; \theta_R) = g(0; \theta_M) = \bar{g}$
→ they must overlap somewhere, or disagreement could not survive
- Signal informativeness (KL divergence):

$$\mathcal{I}_t = \frac{(\theta_R - \theta_M)^2 x_t^2}{2\sigma_\eta^2} \implies \mathcal{I}_t = 0 \text{ at } x_t = 0$$

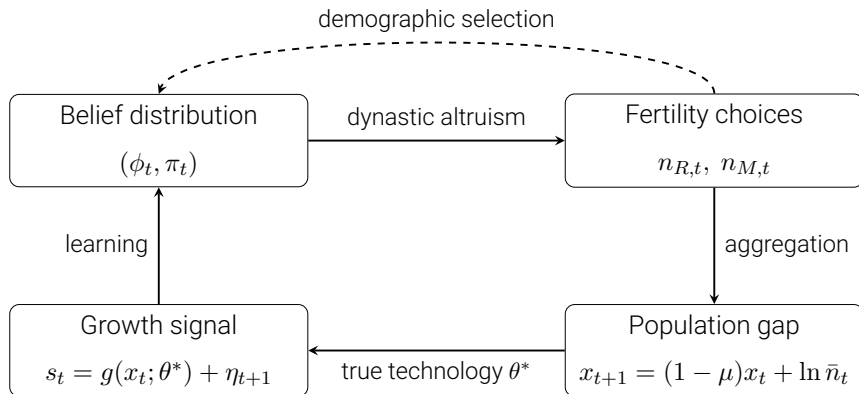
- When $x_t = 0$, observed data cannot help distinguish between worldviews

Households, Beliefs, and Transmission

- Two types: share ϕ_t **Romerians** ($\theta_R > 0$), $1 - \phi_t$ **Malthusians** ($\theta_M < 0$)
- Each acts on its worldview **as if certain**
- **Social credence** π_t : culturally shared posterior, updated by Bayes from growth outcomes $s_t = g(x_t; \theta^*) + \eta_{t+1}$
- Belief distribution evolves via two channels:

$$\phi_{t+1} = \underbrace{\lambda \phi_t^{birth}}_{\text{cultural transmission}} + \underbrace{(1 - \lambda) \pi_{t+1}}_{\text{learning-based recruitment}}$$

The Belief–Fertility–Growth Feedback Loop



- Near $x = 0$: both feedback channels into beliefs shut down together

- Dynastic utility:

$$U_{it} = \frac{c_{it}^{1-\sigma}}{1-\sigma} + \beta n_{it}^{1-\varepsilon} \mathbb{E}_{it}[V_{i,t+1}], \quad \sigma, \varepsilon \in (0, 1)$$

- Time cost $\psi_\sigma(n)$ constructed so marginal cost of a child is **exactly linear**:
 $(1 - \psi_\sigma)^{-\sigma} \psi'_\sigma = \gamma n$
- Consequence: TFP level **cancels** from the fertility FOC
 - level channel (“income supports children”) shut down **by construction**
 - beliefs matter only via **growth forecasts** — where the disagreement lives

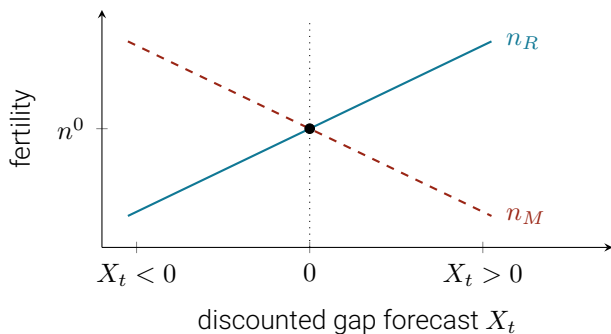
Optimal Fertility Choice

- FOC (TFP cancelled): $\gamma n_j^{1+\varepsilon} = \beta(1 - \varepsilon) \mathbb{E}_j[V'_j]/A^{1-\sigma}$
- Lemma: at the **dormant point** $(x, \bar{n}^e) = (0, 1)$, the FOC is **belief-free**: both types choose the same n^0
- Intuition: beliefs act only on forecasts of the gap
 - gap is zero and expected to stay zero $\Rightarrow$ both types forecast the **same future**
 - disagreement exists but has **nothing to act on**
- Everything a worldview operates on is the **discounted gap forecast**:

$$X_t \equiv \mathbb{E}_t \left[\sum_{k \geq 0} \delta^k x_{t+k} \right] = \frac{x_t + \frac{\delta}{1-\delta} \ln \bar{n}^e}{1 - (1-\mu)\delta}$$

Result 1: State-Dependent Fertility Ranking

$$n_{R,t} - n_{M,t} = K(1 - \sigma)(\theta_R - \theta_M)\bar{w}^0 X_t + O(2)$$



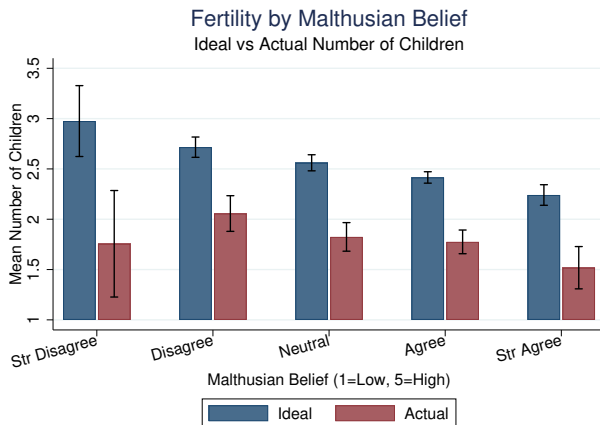
- $X_t > 0$ (growing world): Romerians out-reproduce; $X_t < 0$: ranking flips
- At $X_t = 0$: heterogeneity behaviorally dormant

Correlational Between Belief and Fertility

proxy

regressions

- GSS 2000 & 2010 ($N = 1,822$): agree/disagree “Earth cannot continue to support population growth” – 55% Malthusian
- Malthusians desire 0.27 fewer children (ideal) and have 0.29 fewer (actual)
- Gradient robust to age, education, sex, year, completed fertility

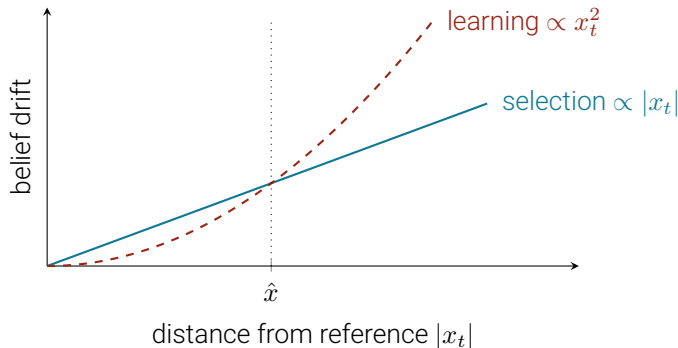


Malthusian Ideas Move High-Stakes Behavior

- Robertson (2026): British Indian officials trained at Haileybury, where **Malthus himself** taught political economy
- Malthus's abrupt death $\Rightarrow$ replaced by Richard Jones (a critic) $\Rightarrow$ quasi-exogenous variation in doctrine across cohorts
- Officials taught by Malthus provided systematically **less famine relief**, decades later
- Three features: exogenous assignment, costly behavior (not cheap talk), **decades-long durability**
- Complements GSS: clean identification of the belief channel, though in bureaucratic relief rather than fertility

- Households atomistic, but optimal fertility depends on forecast of aggregate fertility (via the future gap)
- Finding 1: Romerian fertility is a strategic complement while Malthusian a substitute
- Finding 2: multiplicity requires $\tilde{\theta}_R \gtrsim 10.4$ at baseline $\Rightarrow$ nothing downstream depends much on selection

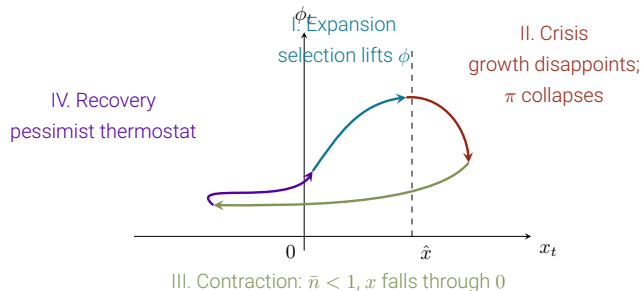
Two Engines of Belief Change



- Selection favors trajectory-optimists – vanishes linearly
- Learning favors the truth – vanishes quadratically
- Near reference: selection dominates; far away: learning dominates; at $x = 0$: both exactly zero

Malthusian World: Boom-Bust Cycles

cycle mechanics



- Selection and learning **oppose** above the reference $\Rightarrow$ tug-of-war
- Busts **reinforced** by selection; recoveries **engineered by pessimists** (they raise fertility when $x < 0$)

Romerian World: The Dormant Trap

details

- At $x = 0$, both views agree on growth $\Rightarrow$ zero info $\Rightarrow$ identical fertility $\Rightarrow$ wrong beliefs can survive forever
- Fertility slightly above replacement $\Rightarrow$ composition slowly shift toward Romerians $\Rightarrow$ trap breaks permanently
- Fertility slightly below replacement $\Rightarrow$ deeper pessimism $\Rightarrow$ signals get quieter $\Rightarrow$ deepening spiral
- Key asymmetry: optimism eventually exposes itself; pessimism hides the evidence that would correct it

Pre-Industrial Era: Informative Depopulation

evidence

- Pre-1800: Malthusian world, $x \approx 0$, Malthusian beliefs dominant $\phi \approx 0$
- The Black Death (1347–51): largest pre-modern demographic shock
 - deep excursion into $x < 0 \Rightarrow$ signal informativeness $\mathcal{I}_t$ high
 - above-trend wage growth observed (Clark, 2007) $\Rightarrow$ strong evidence for θ_M
 $\Rightarrow \pi_t$ collapsed toward Malthusian
- Recovery: Malthusians are high-fertility at $x < 0$ (thermostat) $\Rightarrow$ repopulated, undoing wage gains
- Self-sealing property: every state visited produced either
 - silence at the reference ($x \approx 0$), or
 - confirmation of pessimism (away from zero)
- On the eve of transition, four centuries of evidence pointed the wrong way about the world that was coming

- Puzzle: why the long delay between usable knowledge and takeoff?
- Model: at $x \approx 0$, a Romerian world looks exactly like a Malthusian one
- Two escape channels, both present in the historical literature:
 1. Compositional selection: if $n^0 > 1$, small $x^* > 0 \Rightarrow$ selection slowly favors Romerians $\Rightarrow \phi$ drifts up until the trap destabilizes (Galor–Moav logic)
 2. Cultural shift (Mokyr 2016): Industrial Enlightenment = exogenous surge of optimism about progress, not from macro data but from ideas
- Once escaped, self-cementing: sustained growth keeps data informative, truth locks in

The Modern Fertility Decline: Three Readings

- Fertility below replacement across the developed world
- Three configurations of (θ^*, ϕ_t, x_t) :
 1. **Trap entry**: truth $\theta_R > 0$, but pessimism spreads (overpopulation alarm, climate) $\Rightarrow \bar{n} \rightarrow 1$, entering the deepening spiral from above
 2. **Correct learning**: ideas harder to find (Bloom et al. 2020) $\Rightarrow$ observed weak growth at high x is informative $\Rightarrow$ low fertility is efficient
 3. **Orthogonal forces**: quantity–quality tradeoff dominates; beliefs epiphenomenal (though GSS gradient survives controls)
- Identification: do beliefs track **narratives** (R1) or **productivity news** (R2)?
- Opposite policy implications: pessimism to be fought vs. accommodated

Unconventional Family Policy

welfare

- Planner observes noisy $\hat{\theta}$, announces z with credibility τ_A
- Operates on **beliefs**, not the price of children $\Rightarrow$ closest analogue: **Delphic forward guidance**
- Three channels:
 1. Information provision (shifts π , recruits next generation)
 2. Equilibrium selection (multiplicity regions)
 3. Expectation coordination at the reference: raising expected $\bar{n}$ raises $X_t \Rightarrow$ reactivates the dormant differential and both engines
- Decentralized fertility too low in a Romerian world: scale externality + **head-counting wedge** (planner counts heads, dynasties weight $n^{1-\varepsilon}$)

What to Say, and the Planner's Dilemma

formal results

- Optimal announcement depends on the welfare criterion:

$$\mu_a^{dyn} = \hat{\theta} < \mu_a^{avg} < \mu_a^{tot} = \hat{\theta} + \chi^{tot}$$

- dynastic planner: truthful (unanimity)
- utilitarian planners: optimism shade in Romerian worlds, closing fertility wedges with the only instrument available

The Planner's Dilemma

Public information is most valuable in the trap — but the trap starves the planner of evidence.

Conclusion

- Disagreement about the **sign** of population scale effects generates:
 - state-dependent fertility rankings (validated in GSS)
 - boom-bust cycles in Malthusian worlds
 - an absorbing **dormant trap** in Romerian worlds
- Pessimism delays take-off as optimism self-undermines while pessimism self-protects
- Unconventional family policy, but subject to the planner's dilemma
- Next: causal identification, gradient reversal in depopulating contexts, quantitative implementation

Ecological vs. Economic Malthusianism

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- GSS question is in environment modules: captures **ecological** concern (resource depletion, climate)
- For the fertility margin, distinction is **second-order**: A_t is broad TFP, and both idioms point to $\theta < 0$, a poorer future for descendants
- For **historical** claims the distinction is substantive; the survey only validates the belief-to-fertility arrow

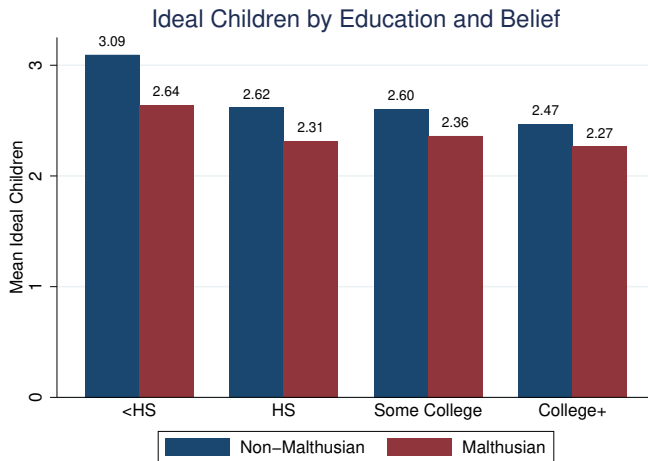
Regression Evidence (Details)

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	Ideal Fertility	Actual Fertility
Malthusian (binary)	−0.268*** (0.041)	−0.292*** (0.066)
Malthusian (1–5 scale)	−0.153*** (0.021)	−0.146*** (0.033)
Controls + Year FE	Yes	Yes
Observations	1,822	1,822

Completed fertility, heterogeneity by education, and further robustness are available in the paper.

Gradient Holds Within Education Groups

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Completed Fertility

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	Full Sample	Completed	Not Yet Completed
Malthusian	−0.292*** (0.066)	−0.257** (0.109)	−0.288*** (0.079)
Observations	1,822	848	974

- Completed = women 45+, men 50+; all controls included
- Estimate stable: not driven by incomplete childbearing among the young

Heterogeneity by Education

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	Ideal		Actual	
	HS or Less	College+	HS or Less	College+
Malthusian	-0.318*** (0.069)	-0.235*** (0.049)	-0.209* (0.115)	-0.392*** (0.081)

- Preferences: gradient across the whole education distribution
- Behavior: stronger among the educated — perhaps fewer constraints on hitting fertility targets

Why Each Preference Ingredient Is Load-Bearing

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- **Time cost + CRRA**: TFP level cancels; income-supports-children channel off by construction
- $\sigma < 1$: required for positive dynastic value of children; mirror $\sigma > 1$ (with $\varepsilon > 1$) flips sign — rejected by data
- $\varepsilon \in (0, 1)$: well-posed fertility problem; becomes the **head-counting wedge** in the welfare analysis
- Boundedness: $\beta \kappa_{\sigma} n_{\max}^{1-\varepsilon} < 1$ ensures finite value; baseline $\beta^{\dagger} \approx 0.22$ from $n^0 = 1$

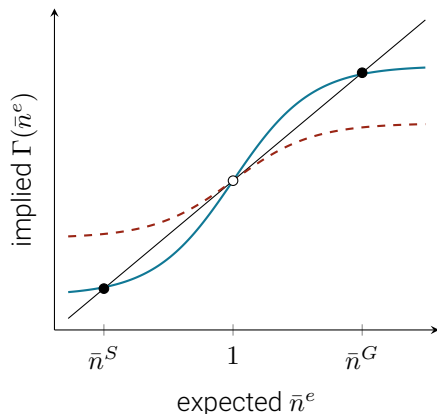
Equilibrium Regions

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Region	Condition	$\rho(\phi)$	Equilibrium
Malthusian-dominated	$\phi < \phi^*$	< 0	Unique
Contraction	$\phi^* \leq \phi \leq \phi^{**}$	$[0, 1)$	Unique
Static multiplicity	$\phi > \phi^{**}, d_R > 1$	> 1	Multiple; RD selection

- $\rho(\phi) = \phi d_R + (1 - \phi)d_M$; pessimists' substitutability enlarges uniqueness regions
- Multiplicity requires $\tilde{\theta}_R \gtrsim 10.4$ at baseline $\Rightarrow$ empty
- Kremer-style calibrations ($\mu \rightarrow 0$): global-games perturbation selects risk-dominant cutoff ϕ^{rd}

Expectational Multiplicity at the Reference

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- $\hat{b}(\phi) < 1$ (dashed): reference determinate; $\hat{b}(\phi) > 1$ (solid): stagnation $\bar{n}^S$ and growth $\bar{n}^G$ flank an unstable reference
- Threshold ordering: trap ($\bar{\theta}_\phi \approx 1.1$) < expectational (≈ 7.7) < static (≈ 10.4)
 $\Rightarrow$ **dynamic** margin is key

Dormant Trap: Formal Statement

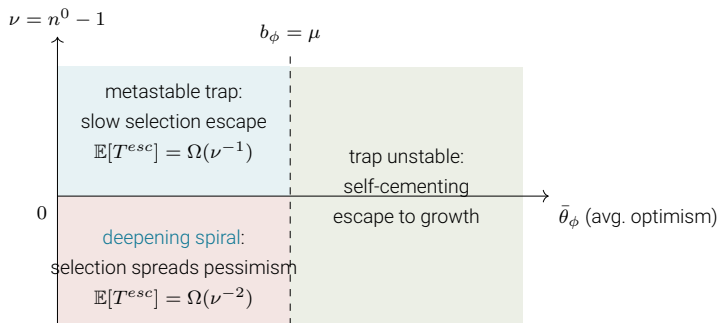
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- Exact dormant point: $x_0 = 0 \Rightarrow x_t = 0, \mathcal{I}_t = 0, \pi_t = \pi_0, n_R = n_M = 1$;
wrong beliefs **never corrected**
- Near-dormant: $|x_t| \leq \rho^t |x_0|, \sum_t \mathcal{I}_t$ finite, $\pi_t \rightarrow \pi_\infty$ with prob. $\geq 1 - C|x_0|/\varsigma$
- Generic $\nu \neq 0$: quasi-steady gap $x^* = \nu/(\mu - \bar{b})$;

$$\mathbb{E}[T^{esc}] = \Omega(|\nu|^{-1}) \text{ if } \nu > 0, \quad \Omega(\nu^{-2}) \text{ if } \nu < 0$$

- **Both** escape engines (selection & learning) vanish with $|x_t|$; stagnation switches them off

Trap Regimes Diagram

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- On event $\{\liminf_t \bar{n}_t \geq 1 + \epsilon\}$: $\liminf x_t > 0$, infinite total information;
 $\pi_t \rightarrow 1, \phi_t \rightarrow 1, g_t \rightarrow \text{Romerian BGP}$
- Complement event has **positive probability** (probability one from deep inside the trap)
- Truth emerges **only along paths that generate evidence**

- Maps onto Beaudry–Galizia–Portier (2020) conditions: strategic complementarities, accumulation variable that depresses further investment (x_t under θ_M), sluggish adjustment (μ)
- Learning strength determines cycle character:
 - **weak** learning: long expansions
 - **strong** learning: quick correction, Malthusian quasi-steady state
 - **intermediate**: pronounced recurrent cycles

- Jones idea production: $\ln(A_{t+1}/A_t) = \nu + \lambda \ln N_t - \omega \ln A_t + \eta_{t+1}$; define x_t to get $\mu = \omega\theta$
- Limit $\mu \rightarrow 0$ reproduces Kremer–Romer strong-scale-effects; trap reverts to slow learning
- **Absorbing** trap is a genuine **semi-endogenous** phenomenon
- θ encodes growth channel net of congestion; level channel neutralized by preferences

Welfare Criteria and Decomposition

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- Three criteria: dynastic, total utilitarian, average utilitarian
- First best vs. decentralized (Romerian world): $n^{FB} > n^{CB}$ from two wedges
 - **scale externality**: no household internalizes children's effect on future gap
 - **head-counting**: planner weights descendants linearly; dynasties weight by $n^{1-\varepsilon}$
- Welfare gap decomposition: Δ_0 (head-count), Δ_1 (externality), Δ_2 (mean bias), Δ_3 (dispersion)
- $\Delta_3 = O(X^2)$: disagreement **costless exactly where it is incorrigible**
- Announcements move Δ_2 and Δ_3 only

- Dynastic: truthful, maximally informative – each type expects vindication; shading is zero-sum
- Total utilitarian: $\chi^{tot} = \frac{\Omega PG}{c + \Omega P^2} > 0$ whenever the fertility wedge is positive; shade **hump-shaped** in pass-through P , peak at $P = \sqrt{c/\Omega}$
- Ordering (Romerian world): $\mu_a^{dyn} = \hat{\theta} < \mu_a^{avg} < \mu_a^{tot}$
- In the trap: $\mathcal{V}^{trap} = pP^{esc}\Delta W^{esc} - (1-p)P^{mis}\Delta W^{mis} - P^{deepen}\Delta W^{deepen}$
 - downside is $O(x_0^2)$; modest precision threshold
 - at high precision, planner's **own noise** binds – formal content of the dilemma

Distinguishing the Historical Interpretations

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- Genuine regime change: centuries of flat real wages (Clark); structural breaks in Unified Growth Theory
- Informational escape: Kremer (1993) shows rising population over millennia; trap reading says scale effect always present, **invisible at $x \approx 0$**
- Trap reading predictions: pessimism favored in below-reference episodes; takeoff preceded by **mortality-driven** net reproduction rise, not productivity acceleration

Escape Mechanics: Why n^0 Matters So Much

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- Dormant-point fertility $n^0 = 1 + \nu$ determines direction of drift
- $\nu > 0$: gap $x^* > 0$, selection slowly favors Romerians; once $b_\phi > \mu$, escape is self-cementing
- $\nu < 0$: gap $x^* < 0$, selection favors [Malthusians](#); false belief spreads, signals get quieter; $\mathbb{E}[T^{esc}] = \Omega(\nu^{-2})$
- Escape trigger is [demographic](#) (n^0 , mortality), not a productivity acceleration

Robustness: What Would Break the Results

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- **No common crossing**: learning always active; trap vanishes
- **No separation**: capped informativeness; crisis and self-cementing escape weaken
- $\sigma > 1$: all rankings reverse, gradient flips — **rejected by data**
- $\mu \rightarrow 0$: trap reverts to slow learning; absorbing character is semi-endogenous
- **Vanishing noise at $x = 0$** : trap dissolves if $\sigma_{\eta}^2(x)$ shrinks as fast as the gap
- **Multiple crossings**: several dormant points; local results survive

Key Thresholds at Baseline

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Threshold	Condition	Baseline value
Trap destabilization	$b_\phi > \mu$	$\bar{\theta}_\phi \gtrsim 1.1$
Expectational indeterminacy	$\hat{b}(\phi) > 1$	$\bar{\theta}_\phi \gtrsim 7.7$
Static multiplicity	$d_R > 1$	$\tilde{\theta}_R \gtrsim 10.4$

- Ordering implies the relevant coordination margin near the reference is the **dynamic** one (trap destabilization)
- Expectational and static thresholds linked by $1/(1 - \delta^0) \approx 1.34$